

Salesforce Developer Course Syllabus

Archer Infotech, Kothrud Pune | Classroom & live-online batches | Updated September 2026

Course Overview

This syllabus lists every module of the course in teaching order, with the topics covered in each. It is generated from the course page, so the two cannot disagree.

Module 1: Platform Foundations for Developers

Schedule: Week 1

The administrator layer a developer cannot skip, compressed into a week. Objects, fields, relationships, record types and the security model — profiles, permission sets, organisation-wide defaults and sharing — because Apex runs in this context and code that ignores it either fails or leaks data.

You set up a Developer Edition org, install the Salesforce Extensions for VS Code, and get a working local development loop before any Apex is written.

Topics

- Objects, fields and relationships
- Record types and page layouts
- Profiles, permission sets and sharing
- Organisation-wide defaults and record access
- Developer Edition org setup
- VS Code and Salesforce Extensions
- Salesforce CLI and authorising an org
- Metadata, source format and the project structure
- Declarative before programmatic — when not to write code

Module 2: Apex Fundamentals

Schedule: Weeks 2–3

The language itself. Data types, collections — lists, sets and maps, with maps given real attention because they are how bulkified Apex avoids queries in loops — control flow, classes, interfaces, inheritance and access modifiers.

sObjects and DML follow: insert, update, upsert, delete, and the difference between the database methods and the DML statements when partial success matters. Exception handling and custom exceptions close the module.

Topics

- Data types, variables and constants
- Lists, sets and maps
- Classes, objects, constructors and properties
- Interfaces, inheritance and access modifiers
- sObjects and field access
- DML statements and Database methods
- Partial success and allOrNone
- Exception handling and custom exceptions
- Apex naming and code conventions

Module 3: SOQL, SOSL & Relationship Queries

Schedule: Week 4

Getting data out of the platform. SOQL syntax, filtering, ordering and limits; parent-to-child and child-to-parent relationship queries, which are the platform's substitute for joins and behave differently from them; aggregate queries and GROUP BY.

SOSL covers text search across objects. The module closes on query plans, selectivity and indexing — the material that decides whether a query survives a million records or times out on them.

Topics

- SOQL syntax, filters and ordering
- Parent-to-child and child-to-parent traversal
- Aggregate queries and GROUP BY
- Date literals and dynamic SOQL
- SOSL for cross-object text search
- Query selectivity and custom indexes
- The query plan tool
- Avoiding queries inside loops
- Binding variables and injection safety

Module 4: Triggers, Bulkification & Governor Limits

Schedule: Week 5

The module that separates Apex that works from Apex that ships. Trigger context variables, before and after events, and the order of execution — which is worth memorising because a surprising number of production defects are explained by it.

Governor limits are covered as a design constraint rather than a list: what they are, why they exist in a multi-tenant platform, and how bulkification, collections and a single trigger per object keep you inside them. You deliberately breach a limit, read the error, and fix the design.

Topics

- Trigger events and context variables
- The order of execution
- One trigger per object and the handler pattern
- Bulkification and processing collections
- Governor limits and why they exist
- Queries and DML outside loops
- Recursion control and static variables
- Trigger frameworks
- Reading and fixing a limit exception

Module 5: Asynchronous Apex

Schedule: Week 6

Work that cannot happen in a synchronous transaction. Future methods for simple offloading and callouts; Queueable for chaining and complex state; Batch Apex for large data volumes with its start, execute and finish structure; and Scheduled Apex for recurring jobs.

The judgement taught is which to reach for, since they have different limits, different debuggability and different failure modes. Platform events and change data capture close the module as the event-driven alternative.

Topics

- Future methods and their constraints
- Queueable Apex and job chaining
- Batch Apex — start, execute, finish
- Batch size, scope and state
- Scheduled Apex and cron expressions
- Choosing between the asynchronous options

- Monitoring jobs and handling failures
- Platform events and event-driven design
- Change Data Capture

Module 6: Lightning Web Components

Schedule: Weeks 7–8

The modern front end. LWC structure — HTML template, JavaScript class, metadata — reactivity, properties and the decorators that drive it. Component composition, events for child-to-parent communication, and the Lightning Message Service for components that are not related.

Data access covers wire adapters, imperative Apex calls, and the Lightning Data Service, with a clear rule for choosing. Navigation, base components, styling with the Salesforce Lightning Design System, and Jest testing close the module.

Topics

- LWC file structure and metadata
- Templates, directives and conditional rendering
- Reactivity, @api, @track and getters
- Component composition and slots
- Custom events and event bubbling
- Lightning Message Service
- Wire adapters versus imperative Apex
- Lightning Data Service and record forms
- Navigation and base components
- SLDS styling and accessibility
- Jest tests for components

Module 7: Integration — REST, SOAP, Callouts & Platform APIs

Schedule: Week 9

Where the salary premium sits. Salesforce as an API provider — REST, SOAP, Bulk and Composite APIs, and when each is right — and as an API consumer: named credentials, external credentials, callouts from Apex, and the callout limits that shape a design.

Authentication covers OAuth flows, connected apps and the JWT bearer flow for server-to-server integration. Error handling, retries and idempotency close the module, along with an honest note on when middleware belongs in the architecture instead of more Apex.

Topics

- REST, SOAP, Bulk and Composite APIs
- Choosing the right API for a volume
- Named credentials and external credentials
- HTTP callouts from Apex
- Callout limits and asynchronous patterns
- OAuth flows and connected apps
- JWT bearer flow for server-to-server
- Apex REST and custom web services
- Error handling, retries and idempotency
- When middleware is the better answer

Module 8: Testing, Debugging & Code Quality

Schedule: Week 10

The platform enforces test coverage before deployment, which turns testing from a discipline into a gate — and produces a great deal of low-value Apex written purely to reach the number. This module covers writing tests that are actually worth having: test data factories, bulk assertions, negative cases, and testing as a user with runAs.

Debugging covers debug logs, log levels, the Apex Replay Debugger and checkpoints. Static analysis, code review and Apex design patterns close the module.

Topics

- Apex test classes and assertions
- Test data factories and @testSetup
- Bulk testing with realistic volumes
- Negative tests and exception assertions
- runAs and testing under different permissions
- Coverage requirements and gaming them
- Debug logs, levels and reading a log
- Apex Replay Debugger and checkpoints
- Static analysis and code review
- Common Apex design patterns

Module 9: Deployment, Salesforce DX & DevOps

Schedule: Week 11

Getting work from your org to production safely. Sandboxes and their refresh behaviour; change sets and their limits; then Salesforce DX properly — source-driven development, scratch orgs, the project structure, and unlocked packages.

Version control with Git, branching strategy for Salesforce teams, and a CI pipeline that validates a deployment before it runs. The module closes on release management across the three annual Salesforce releases, which is a real operational obligation rather than a footnote.

Topics

- Sandbox types and refresh behaviour
- Change sets and where they run out
- Salesforce DX and source-driven development
- Scratch orgs and org shape
- Unlocked and managed packages
- Git branching for Salesforce teams
- CI pipelines and validation deployments
- Destructive changes and metadata dependencies
- The three annual releases and impact assessment

Module 10: Capstone Build & Platform Developer I Preparation

Schedule: Weeks 12–13 + placement prep

A full build in your own org — data model, triggers with a handler framework, asynchronous processing, Lightning Web Components, an integration and a test suite — deployed through a pipeline you configured, with a README explaining the architecture.

Alongside it, structured preparation for the Platform Developer I exam: section weightings, scenario question style, timed mock papers and systematic work through wrong answers. Mock interviews cover Apex, LWC, governor limits and integration scenarios.

Topics

- Capstone implementation and deployment
- Architecture documentation and README
- Platform Developer I exam weightings

- Scenario question technique and pacing
- Timed mock papers and gap closure
- Mock interview — Apex and governor limits
- Mock interview — LWC and integration
- Walking a panel through your own code
- Resume and LinkedIn for developer roles
- Booking, proctoring and maintenance

Module 11: AI-Assisted Salesforce Development & Agentforce

Schedule: Final Week

AI in the development loop, and where Agentforce fits for a developer. Using assistants to scaffold Apex and LWC, generate test classes, explain inherited code and draft documentation — with every suggestion reviewed, because a hallucinated method on a standard object compiles in the assistant's imagination and not in your org.

Agentforce is covered from the developer side: agent actions backed by Apex, grounding on org data, and the integration work that feeds an agent something worth reasoning over. The career guide's caveat applies here too — this sits on top of platform fundamentals, not instead of them.

Topics

- AI assistants for Apex and LWC scaffolding
- Generating test classes and improving coverage
- Explaining and refactoring inherited Apex
- Spotting hallucinated methods and objects
- Documentation and code-comment generation
- Agentforce agent actions backed by Apex
- Grounding an agent on org data
- Integration as the input to useful AI
- Data privacy when using external AI tools
- A mini-project built with an AI-assisted workflow

Enquire or book a free demo session. Archer Infotech, Flat No. 12, Divyadarshan Housing Society, behind Kothrud Bus Stand Road, Kothrud, Pune 411038. Phone +91 9850 678451. Email info@archerinfotech.in. Web archerinfotech.in.